Everything these two chapters ask you to be able to do.

Each competency is written out in full, with two brain dump prompts underneath it. Pick one prompt per competency and do that one on your note sheet. You do not need to open any other list.

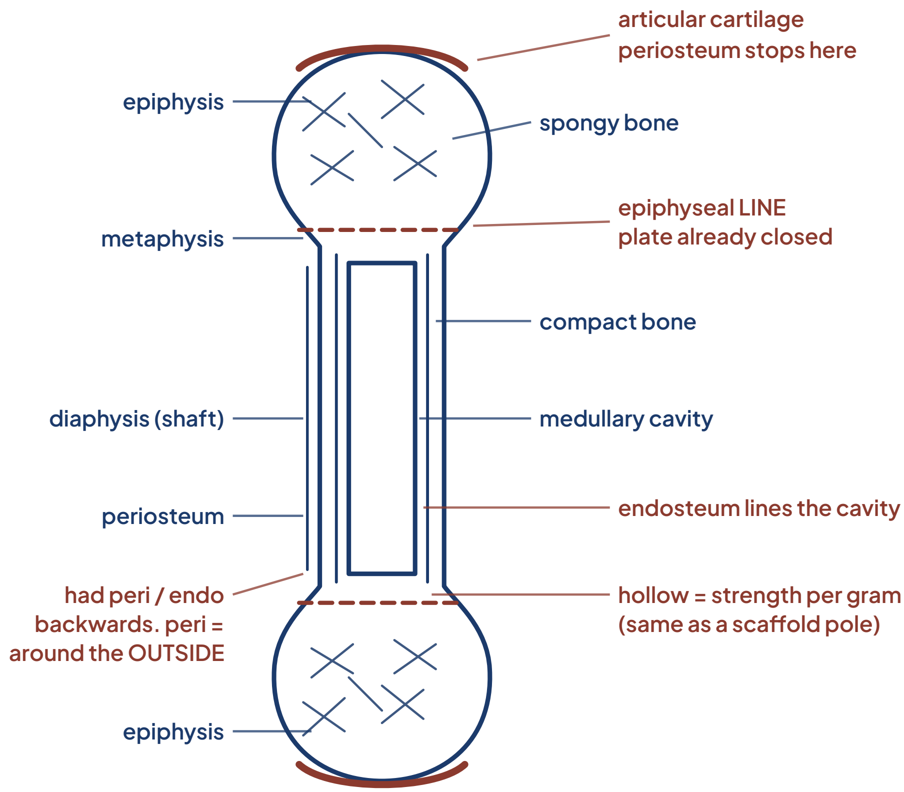
What a finished box looks like on your paper sheet

These two are **anatomy**, from BIO 004, on purpose. Nothing here goes on your physiology sheet, so you are looking at how a box is built and not at answers to copy. Draw, do not write paragraphs. One concept gets one box, and when you move to the next concept you start a new box.

4. Long bone structure

Prompt A · Draw a long bone in section and label every region

- [ ] FIRST PASS, BEFORE THE VIDEO
- [ ] SECOND PASS, AFTER THE VIDEO

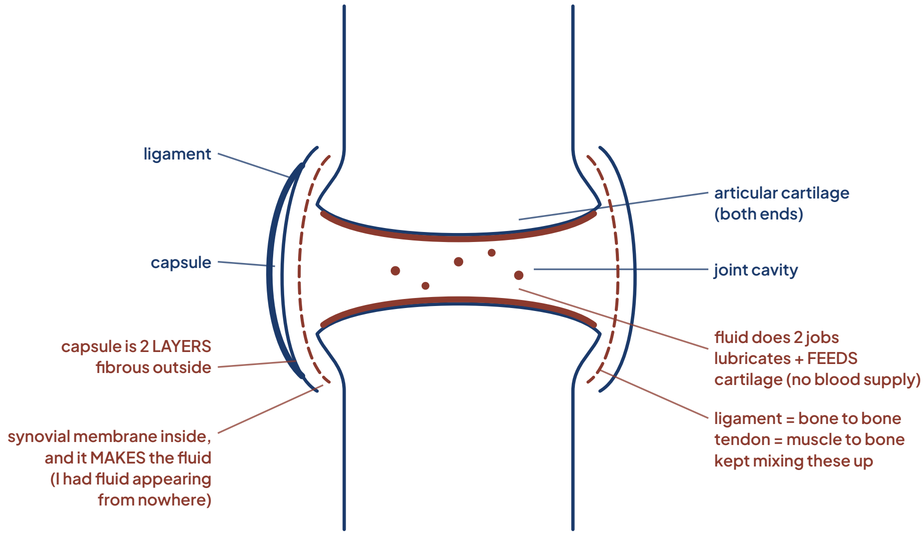


NEW CONCEPT, NEW BOX ↓

5. Synovial joint structure

Prompt B · List the parts every synovial joint has and what each one does

- [ ] FIRST PASS, BEFORE THE VIDEO
- [ ] SECOND PASS, AFTER THE VIDEO



Notice what makes these gradeable. The header names the competency and says which prompt was chosen. The first pass is a real drawing with labels, not a paragraph, and it leaves white space. The second pass goes into the same drawing, adds what was missing and says what was wrong, not just what is right. That gap between the two colours is the first thing I look at.

SILVERTHORN CHAPTER 1

Introduction to physiology

What physiology asks, homeostasis, mass balance, the reflex pathway, and the two kinds of feedback.

1 Physiology and levels of function (LECTURE)

YOU SHOULD BE ABLE TO

Define physiology and place a given process at the correct level of organization from molecule to organism, then state the level at which that process is best explained.

WORKED EXAMPLE, SO YOU CAN SEE WHAT A FINISHED ONE LOOKS LIKE

Carrying oxygen, one case all the way up the ladder

Do not stop at naming the levels. Pick one thing your body does and follow it up, so each rung is the same story at a bigger scale.

MOLECULE	Hemoglobin. Four subunits, each one binds a molecule of oxygen.
CELL	The red blood cell. Packed with hemoglobin, and it drops its nucleus to make room for more.
TISSUE	Blood. Red cells suspended in plasma, moving as one fluid tissue.
ORGAN	The lung loads the oxygen on. The heart pushes the loaded blood out.
ORGAN SYSTEM	Respiratory and cardiovascular, working as one delivery line.
ORGANISM	You climb the stairs without stopping.

Now break one rung and follow the damage up

Change one amino acid at the bottom and every level above it changes too. That is how you know the levels are real and not just labels.

MOLECULE	In sickle hemoglobin, valine sits at position 6 of the beta chain where glutamic acid should be. Once it gives up its oxygen, the molecules stick to each other and build long polymers.
CELL	Those polymers push the red cell out of shape into a stiff sickle. It no longer bends to get through a capillary.
TISSUE	Sickled cells are destroyed far faster than the marrow can replace them, so the blood carries less oxygen. The stiff ones also jam in small vessels and stop flow.
ORGAN	Downstream tissue is starved of oxygen. That is what a pain crisis is, and over years it damages the spleen, the kidneys, the brain.
ORGAN SYSTEM	The delivery line runs short on both counts, less oxygen carried and blocked routes to deliver it through.
ORGANISM	Fatigue, breathlessness, episodes of severe pain.

Your turn

Do not use oxygen delivery. Pick your own process and build both halves for it, the ladder up and the break. Anything works as long as you can follow it through all six levels: contracting a muscle, digesting a meal, clotting a cut, hearing a sound, filtering blood in the kidney. Choose the one you are most curious about, because you will meet it again later in the term.

PICK ONE AND DO THAT ONE

- [ ] **A** Build your own ladder. Pick a process, not oxygen delivery, and draw it at all six levels, molecule up to organism. One small box per level, labeled. Then break one rung and write, level by level, what stops working above it.

- [ ] **B** Pick your own process, not oxygen delivery. Draw the same thing twice. First as anatomy sees it, the structures and their parts. Then as physiology sees it, the same thing working, with arrows for what moves and what changes. Under each drawing, one line naming the question it answers.

2 Structure and function relationship LECTURE LAB

YOU SHOULD BE ABLE TO

Predict how a change in the structure of a molecule, cell, tissue, or organ alters its function, using examples from two different organ systems.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw one structure from this week and label the single feature that makes its job possible. Then sketch what would stop working if that feature changed.
- [ ] **B** List three structures from this week. Beside each, write the one property its function depends on, in six words or fewer.

3 Body fluid compartments LECTURE LAB

YOU SHOULD BE ABLE TO

State the approximate volumes of total body water, intracellular fluid, extracellular fluid, plasma, and interstitial fluid in a 70 kg adult and compare the dominant solutes of the intracellular and extracellular compartments.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw the fluid compartments top down with the fractions. Use one border style for inside cells and a different one for outside.
- [ ] **B** On a compartment drawing, mark where a blood sample is taken and which compartment you are actually drawing a conclusion about.

4 Compartment separation and clinical volume shifts LECTURE

YOU SHOULD BE ABLE TO

Predict the direction of water movement between compartments when extracellular osmolarity rises or falls and name a clinical situation that produces each shift.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw a patient with swollen ankles and a low blood pressure. Use arrows to show where the water went.
- [ ] **B** Draw three scenarios: water gained in the vessels, water lost from the body, water moved into tissue. Predict blood pressure under each.

5 Homeostasis defined LECTURE

YOU SHOULD BE ABLE TO

Define homeostasis, regulated variable, and setpoint, and distinguish homeostasis from chemical equilibrium and from steady state.

PICK ONE AND DO THAT ONE

- [ ] **A** Write the definition of homeostasis, then underline the three words doing the real work. Beside each underlined word, one line on why it matters.
- [ ] **B** Draw two graphs side by side, a value in steady state and a value at equilibrium. On each, label what is still moving and what it costs.

6 Mass balance LECTURE LAB

YOU SHOULD BE ABLE TO

Apply the mass balance equation to a solute or to body water and calculate the intake, production, and output combination that holds the amount in the body constant.

PICK ONE AND DO THAT ONE

- [ ] **A** Write the mass balance equation and label every term. Then draw a body outline with arrows in and out for sodium, labeling which arrow is which term.
- [ ] **B** Work a mass balance for one substance across one day. Then change a single term and show the new result underneath.

7 Mass flow and clearance LECTURE

YOU SHOULD BE ABLE TO

Calculate mass flow as concentration times volume flow, and clearance as rate of removal divided by plasma concentration, and explain why two patients with the same concentration can have very different delivery.

PICK ONE AND DO THAT ONE

- [ ] **A** Write both equations from memory with the units written under every term, and show the units cancelling. Then work the two patient case: identical oxygen content, one with half the cardiac output. Numbers, not sentences.
- [ ] **B** Draw one vessel delivering to a tissue. Label the concentration and the volume flow. Draw it again with the flow halved and write what happened to delivery. Then say in one line which one you would change first in a patient, and why.

8 Feedback loop components LECTURE LAB

YOU SHOULD BE ABLE TO

Diagram a negative feedback loop labeling stimulus, sensor, afferent path, integrating center, efferent path, effector, and response, and trace body temperature or blood glucose through every step.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw the five component reflex pathway top down and label all five plus the feedback arrow. Labels only, no sentences.
- [ ] **B** Fill in the five components for thermoregulation on a cold day. Then mark which component you would suspect first if the patient never shivered.

9 Local and reflex control pathways LECTURE

YOU SHOULD BE ABLE TO

Distinguish a local control pathway from a long distance reflex pathway and classify a given response as one or the other.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw a local control example and a reflex control example side by side. On each, mark how far the signal travels.
- [ ] **B** Standing up suddenly: draw the reflex, label all five components, and write one line on why local control could not have handled it.

10 Negative and positive feedback `LECTURE`

YOU SHOULD BE ABLE TO

Distinguish negative from positive feedback by the direction of the response and give a physiological example of each, explaining why a positive feedback loop needs an outside event to end it.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw one negative feedback loop and one positive feedback loop as arrow diagrams. On each, mark what makes it stop.
- [ ] **B** Draw clotting as a positive feedback loop, then show what ends it. Underneath, write one line on why the body can afford to use positive feedback here.

11 Feedforward control and acclimatization `LECTURE`

YOU SHOULD BE ABLE TO

Explain anticipatory feedforward control and acclimatization and identify which one is operating in a given scenario.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw a timeline of eating a meal. Mark where feedforward starts and where feedback takes over.
- [ ] **B** Draw a setpoint as a horizontal line. Show what fever does to it, and mark at each stage what the patient feels.

SILVERTHORN CHAPTER 6

How signals travel

The four routes cells use to reach each other, where a receptor sits and why, how one bound molecule becomes a large response, and how a cell changes its own sensitivity.

12 Signal types and range `LECTURE`

YOU SHOULD BE ABLE TO

Classify a chemical signal as autocrine, paracrine, neurotransmitter, neurohormone, or hormone by its route and distance of travel.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw one target cell in the middle of the box. Draw an arrow in from each of the five signal types, autocrine, paracrine, neurotransmitter, neurohormone, hormone. Label each arrow with how far it travelled and what carried it.
- [ ] **B** Make a five row table. Signal type, where it is released, how far it goes, what it travels through. Fill it from memory, then check one row against the book.

13 Receptor location and ligand solubility `LECTURE`

YOU SHOULD BE ABLE TO

Predict whether a signal molecule binds a surface receptor or an intracellular receptor from its lipid solubility and relate that to the speed and duration of the response.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw a cell with one surface receptor and one intracellular receptor. Show the path a lipid soluble signal takes and the path a water soluble one takes. Mark which response is faster and which lasts longer.
- [ ] **B** Write the two questions you would ask about any new signal molecule to predict where its receptor sits. Then answer both for a steroid and for a peptide hormone.

14 Signal amplification LECTURE

YOU SHOULD BE ABLE TO

Explain how a cascade amplifies a signal and estimate the size of the amplification across the steps of a given pathway.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw a three step cascade as a widening funnel. Put a number on each step and multiply them down the side to get the total amplification.
- [ ] **B** Start from one ligand and write the count at each step of a cascade until you reach the response. One line per step, numbers only, no sentences.

15 Receptor modulation LECTURE

YOU SHOULD BE ABLE TO

Define agonist, antagonist, competitive inhibition, up regulation, and down regulation, and predict the effect of chronic ligand excess on target cell sensitivity.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw one target cell three times. Normal, after a long stretch of scarce signal, after a long stretch of excess signal. Show the receptor count changing in each and label which is up regulation and which is down.
- [ ] **B** Define agonist, antagonist, up regulation and down regulation in your own words, one line each. Then explain in two lines why a patient who needs more opioid for the same effect is usually showing down regulation rather than drug seeking.

CONCEPT 13 AND THE LAB

Doing physiology as a science

How a physiology claim gets tested, and how you tell a real difference from noise.

16 Experimental design and controls LECTURE LAB

YOU SHOULD BE ABLE TO

Identify the hypothesis, independent variable, dependent variable, and control condition in a physiology experiment and state what the control rules out.

PICK ONE AND DO THAT ONE

- [ ] **A** Sketch an experiment as three labeled boxes: what you changed, what you measured, and your control.
- [ ] **B** Take a claim from this week and design the control group that would test it. Draw both groups and label the one difference between them.

17 Measurement error and variability LECTURE LAB

YOU SHOULD BE ABLE TO

Distinguish random from systematic error and explain why physiological measurements are repeated and averaged.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw two sets of repeated measurements, one with high variability and one with low. Mark which difference between readings is real.
- [ ] **B** List three sources of variation in a single measurement. Beside each, one way to reduce it.

18 Graphing and data interpretation

LECTURE LAB

YOU SHOULD BE ABLE TO

Construct a labeled graph with the independent variable on the x axis, and read slope, direction, and trend from a physiological data set.

PICK ONE AND DO THAT ONE

- [ ] **A** Plot the mission patient's sodium across the three days. Label both axes with units and mark the reference range on the graph.
- [ ] **B** Draw a saturation curve. Label the plateau, then write one line on what the plateau means physiologically.